

Week 4 — Variable roles: confounding & DAGs

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES

Today's objectives

- ▶ Name the **four roles** a covariate can play — confounder, mediator, collider, risk marker — and the adjustment rule for each
- ▶ Draw a **minimal DAG** for the MASLD question and read a backdoor path off it
- ▶ Explain why **age** is the dominant confounder, using the ~40 vs ~62-year group age gap
- ▶ Use **Table 1** for what it can do — **describe** the groups, expose **sparse cells**, **positivity** problems and **coding errors** — not to **identify confounders**; and read it critically enough to catch a mislabelled cut point
- ▶ Interpret the **crude** → **adjusted collapse** of the Group IV hazard ratio **without claiming which mechanism produced it**, and read the “obesity paradox” as a **structural question**, not a settled artefact

Where we've been, where we're going

- ▶ **Last week (Wk3 → L2 / P1):** we locked the analytic cohort — eligibility, the **FLI 60** exposure filter, the frozen **N**. Milestone **M0 is closed**; your paper is locked.
- ▶ **This week:** with the cohort fixed, we ask what *distorts* the exposure–mortality contrast — variable roles, the DAG, and Table 1 as a description of the groups that raises questions the causal argument has to answer.
- ▶ **Sets up:** Thursday's **L3** takes Table 1 → crude → adjusted; **P2** is your own Table 1 plus your adjustment rationale.

MASLD: the cohort is settled — now we ask whether Group IV's mortality gap is *real or confounded*.

Slide 1 — Where we are in the project

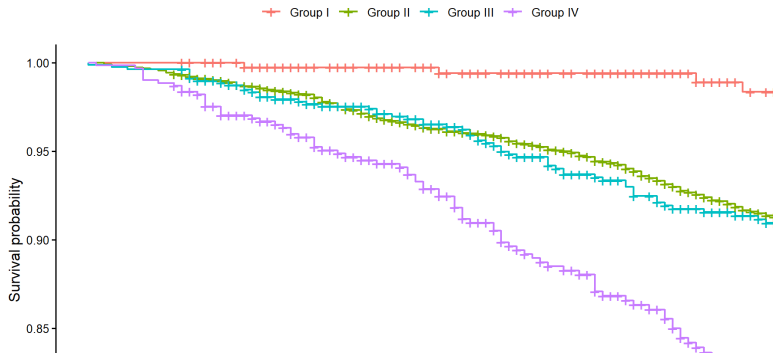
- ▶ The arc: **Replicate** → **Interrogate** → **Improve** → **Defend**
— we are entering **Interrogate**
- ▶ **M0 locked** last week: the cohort and the four phenotype groups are now fixed inputs — the paper reports **N 6,300**, our **full analytic file** is **6,371**
- ▶ Today is purely **conceptual scaffolding** for adjustment — no new data steps
- ▶ Groups are settled — paper: **I 386 / II 4,541 / III 769 / IV 604**; ours: **I 391 / II 4,585 / III 787 / IV 608**

MASLD: the four groups are fixed — today asks whether their mortality contrast is real or confounded.

Slide 2 — The motivating puzzle

- ▶ Group IV (**non-obese, high central adiposity**) has the **worst survival** — but the number moves enormously with adjustment
- ▶ Unadjusted all-cause **HR 15.1**; adjusted **2.9–3.3** — same data, one covariate set apart
- ▶ Ask the room: *what could make a hazard ratio shrink five-fold?* Park the answers — we return at Slide 13

All-Cause Mortality



Slide 3 — Four roles a variable can play

- ▶ A covariate is **not “a control” by default** — its role depends on the causal structure, not on the regression
- ▶ **Confounder** — common cause of exposure and outcome
- ▶ **Mediator** — sits *on* the causal path
- ▶ **Collider** — a common *effect*
- ▶ **Risk marker** — predicts the outcome, no exposure link
- ▶ The role dictates the action: **adjust confounders**; *don't* adjust mediators or colliders

MASLD: age, the cardiometabolic criteria, and the FLI filter each play *different* roles — naming them is the task.

Slide 4 — Confounding: a common cause

- ▶ **Definition:** a variable that causes *both* exposure and outcome — opening a **backdoor path**
- ▶ **Signature:** it **causes both** the exposure and the outcome. Imbalance + prognosis tell you **where to look**, not what the variable **is**
- ▶ “Adjusting it changes the estimate” is **not** a test — mediators and colliders move it too, and so do **non-collapsibility** and a **changed sample**
- ▶ This is the **one** role where adjustment *removes* bias
- ▶ Rule: **close** the backdoor by conditioning on the common cause

MASLD: age raises central adiposity / phenotype membership *and* raises mortality — the textbook confounder here.

Slide 5 — Mediation: on the causal path

- ▶ **Definition:** exposure $\rightarrow$ **mediator** $\rightarrow$ outcome; the mediator *transmits* the effect
- ▶ Adjusting a mediator removes part of the **very effect you want** — “**over-adjustment**”
- ▶ Same regression syntax as confounding — **opposite** consequence
- ▶ Diagnostic question: is this variable *downstream* of the exposure?

MASLD: if central adiposity drives mortality *through* incident diabetes / CVD, controlling those downstream conditions would **erase real effect** — a mediation trap to flag.

Slide 6 — Collider bias: conditioning on a common effect

- ▶ **Definition:** exposure $\rightarrow$ **C** $\leftarrow$ outcome (or their causes); **C** is a collider on that path
- ▶ Conditioning on a collider — by **adjustment** or by **selection** — *opens* a spurious path
- ▶ **Selection into the sample is a form of conditioning** — who gets measured matters
- ▶ Lesson: don't reflexively “adjust for everything”

MASLD: the paper adjusts for T2DM, hypertension, CKD and prior MI — ask which the class can **justify** as confounders rather than merely observe to be prognostic. Several could be **descendants or intermediates** under plausible causal structures, and the data do not settle their causal roles or temporal ordering.

Slide 7 — A minimal DAG for the MASLD question

- ▶ Draw live: **Phenotype** (exposure) → **Mortality** (outcome)
- ▶ **Age** → **Phenotype** and **Age** → **Mortality** — a **backdoor path**
- ▶ **Cardiometabolic factors** as shared causes on the same footing as age
- ▶ Trace the open backdoor age creates; show that **adjusting Age blocks it**
- ▶ Add a mediator arrow — **Phenotype** → **diabetes** → **Mortality** — to mark what *not* to block

MASLD: one small hand-drawn DAG makes visible *why age is adjusted* and *why downstream disease is not*.

Slide 8 — Age: the dominant confounder

- ▶ Group mean ages differ **sharply** — roughly ~**40** in the younger obese groups vs ~**62** in the leaner high-central group
- ▶ Non-obese high-central adults are, on average, **older** — and age alone predicts death
- ▶ So a chunk of Group IV's raw excess mortality is “**being older,**” not the phenotype
- ▶ This is the confounder signature from Slide 4, made concrete

MASLD: the ~**40** vs ~**62** gap is the single largest driver of the crude **HR 15**.

Slide 9 — The phenotype definition (exposure construction)

- ▶ Two axes: **obesity (BMI ≥ 30) × central adiposity (WHtR ≥ 0.6)** → four groups (I–IV)
- ▶ **Group IV = non-obese (BMI < 30) but high central adiposity** — the “**hidden risk**” cell
- ▶ The **cut points are analyst choices** — they determine who lands in the worst-survival group
- ▶ Change the cut, change the group — the exposure is *constructed*, not given

MASLD: Group IV’s definition is exactly why “non-obese” can still be high-risk — the paper’s headline.

Slide 10 — Reading tables critically: the mislabelled cut

- ▶ The published Table headers print the BMI cut as “< 25” when the actual analytic cut is **30**
- ▶ A reader trusting the header would **misread the entire exposure** — the *code*, not the label, is truth
- ▶ **Lesson:** reconcile every table label against the methods and the derivation
- ▶ Replication catches what careful reading alone cannot

MASLD: this is a **real, reproducible label–Methods mismatch in Kueh 2026** — a live example of why replication beats reading.

Slide 11 — BREAK (10 min)

- ▶ Stretch and reset
- ▶ **Leave the DAG on the board** for the second half

Slide 12 — What Table 1 can and cannot tell you

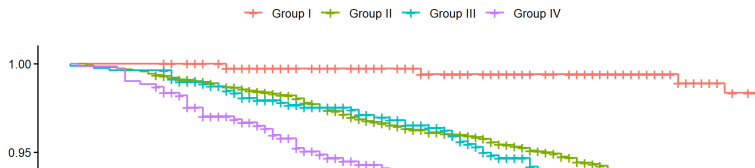
- ▶ Reframe Table 1: **not decoration**, but the **description** of how the groups actually differ — a **data** diagnostic, not a **confounder detector**
- ▶ Read **across** the phenotype columns — age, sex, cardiometabolic markers — looking for **sparse cells**, **positivity** problems (a covariate level with almost nobody in one group), **impossible or miscoded values**, and contrasts *worth investigating*
- ▶ **Imbalance decides nothing** — the adjustment set comes from **causal and temporal reasoning**; a badly imbalanced variable can still be a **mediator** or a **collider**, and the set is justified from **science, not a stepwise algorithm**
- ▶ Table 1 reproduces closely: mean **age** ~51, **BMI** ~33.2

MASLD: the **age row** (~40 vs ~62) is the *question* Table 1 hands you — whether age **confounds** this contrast is a claim about **causal structure**, argued on the **Slide 7 DAG**, not read off the row.

Slide 13 — Crude vs adjusted: the collapse explained

- ▶ Return to Slide 2: Group IV all-cause **HR 15 crude** → **3.3 after adjustment**
- ▶ Mechanism: adjustment can **remove confounding**, **remove real effect** through mediators, or **add bias** through colliders
- ▶ The hazard ratio is also **non-collapsible** — it moves on adjustment even under a null — and crude and adjusted must be fitted on the **same rows**
- ▶ **Attenuation is diagnostic evidence about what the covariates are doing** — not evidence that the adjusted estimate is correct
- ▶ The residual **3** is the **remaining conditional association under this adjustment set** — not the phenotype's own signal

All-Cause Mortality



Slide 14 — The “obesity paradox” as a structural question

- ▶ The paradox: obese patients sometimes look **protected** in crude comparisons
- ▶ The candidate explanations are **structural, not biological** — confounding by age / severity, selection into a sick-but-surviving sample — but **naming a candidate is not demonstrating one**
- ▶ **Descriptively**, “leaner is safer” fails and the paper reports that accurately: Group IV has the **poorest observed survival** — **19.5%** died, against **9.6%** in Group III
- ▶ **Comparatively**, it does not follow: on the locked **IV-vs-III** contrast the adjusted HR is **1.06 (0.77–1.45)** unweighted, and the paper’s own Model 1 puts Groups III and IV almost on top of each other — implied ratio about **0.99**
- ▶ A DAG **frames** the paradox; it does not **settle** it

MASLD: the headline is **not** the paradox resolved — it is the paradox **posed** on the cell that holds body-mass category fixed.

Wk6 re-runs that contrast design-aware: **1.29 (0.90–1.86)**, interval still including 1

Slide 15 — What adjustment can and cannot fix

- ▶ Adjustment closes **measured** backdoor paths only — **residual / unmeasured** confounding remains
- ▶ Over-adjustment (mediators) and collider / selection bias can *add* bias — **more covariates is not always better**
- ▶ Preview: **survey design (Wk6)** and **missing-FLI selection (Wk7)** are biases adjustment alone won't fix
- ▶ The adjusted **3 is provisional** — later weeks stress-test it

MASLD: even the adjusted 3 is not the last word — the survey weights and the FLI selection are still ahead.

Slide 16 — Bridge to today's lab

- ▶ **L3 (Thursday):** in the **EpiMethods OER** environment, move from the frozen cohort to estimates — build **Table 1** across the four phenotype groups, then fit **crude and age-adjusted** models to watch the Group IV **HR fall from 15 toward 3**
- ▶ **Workflow:** use Table 1 to **see the groups** and catch **sparse cells, positivity and coding problems**; decide the adjustment set from **causal and temporal reasoning** (today's DAG logic); **record what you deliberately leave out** (mediators, colliders)
- ▶ **P2 (your paper):** produce **your own Table 1** plus a written **adjustment rationale** — which variables you adjust, which you don't, and why — carrying forward into your **M1** replication

MASLD: you will reproduce, with your own hands, the **crude → adjusted collapse** we drew today.